

Conic Reformulation of Quasi-Linear EG Clearing

Interior Point Solving via Exponential Cones

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Summary. The Eisenberg-Gale clearing problem — a linear program augmented with K logarithmic utility terms — admits two efficient interior point formulations. The *conic reformulation* introduces K exponential cone constraints and hands off to Clarabel.rs, achieving a clean single-solve approach. The *augmented LP* reformulation reveals a deeper insight: the EG Hessian is a rank- K perturbation of the LP’s diagonal barrier, so Newton’s method on the EG problem has the same per-iteration cost as LP interior point. Both replace the current Frank-Wolfe approach (25 LP oracle calls) with a single interior-point solve, targeting 10–25× speedup.

1. The Quasi-Linear EG Program

The risk-averse batch clearing program (companion paper, Theorem 5):

$$P^{\text{RA}} : \max_{\mathbf{q} \geq 0, \mathbf{s} \geq 0} \underbrace{\sum_k [B_k \ln(V_k) - s_k]}_{\text{MM log utility}} + \underbrace{\sum_{j \notin \text{MM}} w_j q_j}_{\text{retail welfare}} - \underbrace{N \cdot \sum_m \text{mint}_m + N \cdot \sum_g \text{gmint}_g}_{\text{minting cost}}$$

where:

- $V_k = U_k + s_k = \sum_{i \in \text{MM}_k} L_i q_i + s_k$ is MM k ’s total value (fill utility plus retained cash)
- $L_i = \text{sign}_i \cdot \text{limit}_i$ is the welfare weight of order i
- $N = 10^9$ (nanos per dollar)
- $s_k \geq 0$ is cash retained by MM k (quasi-linear extension)

Subject to:

- **Position balance** (per market m):

$$\sum_i c_{y,i,m} q_i = \text{mint}_m + \text{gmint}_{g(m)}, \quad \sum_i c_{n,i,m} q_i = \text{mint}_m$$

- **Box constraints:** $0 \leq q_i \leq \bar{q}_i$, $s_k \geq 0$, $\text{gmint}_g \geq 0$, mint_m free.

Observation 1 (structure). The objective is concave: $B_k \ln(V_k)$ is concave in $(\mathbf{q}, s_k)$ (perspective of $\ln$ preserves concavity). All constraints are linear. This is a *linearly constrained concave maximization* — the natural domain of interior point methods.

Observation 2 (LP reduction). Without the $\ln$ terms (i.e., $B_k = 0$ for all k , or equivalently no MMs), the program reduces to the LP solved by `lp_solver.rs`. The K log terms are the *only* source of nonlinearity.

Observation 3 (cash variable). The KKT condition for s_k gives $B_k/V_k \leq 1$, with equality when $s_k = 0$. At optimum:

$$V_k = \max(U_k, B_k)$$

Over-capitalized MMs park $s_k = B_k - U_k > 0$ as cash, preventing budget distortion of fills. The program automatically recovers LP behavior when budgets don’t bind.

2. Conic Reformulation (Clarabel)

Clarabel solves conic programs in standard form:

$$\min_{\mathbf{x}} \quad \frac{1}{2} \mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{c}^T \mathbf{x} \quad \text{s.t.} \quad \mathbf{A} \mathbf{x} + \mathbf{s} = \mathbf{b}, \quad \mathbf{s} \in \mathcal{K}$$

where $\mathcal{K} = K_1 \times K_2 \times \dots$ is a Cartesian product of supported cones.

2.1. Variables

The decision vector $\mathbf{x} \in \mathbb{R}^d$ with $d = n + K + K + M + G$:

Symbol	Meaning	Count	Bounds
q_i	Fill quantity, order i	n	$[0, \bar{q}_i]$
s_k	Retained cash, MM k	K	≥ 0
t_k	Log-utility epigraph	K	free
mint_m	Per-market minting	M	free
gmint_g	Group minting	G	≥ 0

2.2. Objective

Minimize (negated welfare):

$$\mathbf{c}^T \mathbf{x} = - \sum_k B_k t_k + \sum_k s_k - \sum_{j \notin \text{MM}} w_j q_j + N \cdot \sum_m \text{mint}_m + N \cdot \sum_g \text{gmint}_g$$

No quadratic term: $P = 0$. The curvature lives entirely in the cone constraints.

2.3. Exponential Cone Constraints

For each MM k , the constraint $t_k \leq \ln(V_k)$ is modeled via the exponential cone:

$$\mathcal{K}_{\text{exp}} = \{(x, y, z) : y \exp(x/y) \leq z, \quad y \geq 0\}$$

Setting $(s_1, s_2, s_3) = (t_k, 1, V_k)$ gives $\exp(t_k) \leq V_k$, i.e., $t_k \leq \ln V_k$. ✓

In Clarabel's slack form $\mathbf{A} \mathbf{x} + \mathbf{s} = \mathbf{b}$, each MM k produces three rows:

$$\begin{pmatrix} -\mathbf{e}_{t_k} \\ \mathbf{0} \\ -\mathbf{L}_k - \mathbf{e}_{s_k} \end{pmatrix} \mathbf{x} + \begin{pmatrix} s_1 \\ s_2 \\ s_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

where $\mathbf{e}_{t_k}$ selects the t_k variable and $\mathbf{L}_k$ has L_i at position i for each $i \in \text{MM}_k$.

2.4. Linear Constraints

Position balance (zero cone): For each market m , two equality rows. $2M$ rows total.

Box constraints (nonnegative cone): Upper bounds $\bar{q}_i - q_i \geq 0$, lower bounds $q_i \geq 0$, cash $s_k \geq 0$, group minting $\text{gmint}_g \geq 0$. Also mint_m is free (no cone constraint).

2.5. Full Cone Specification

$$\mathcal{K} = \underbrace{\mathcal{K}_{\text{exp}}^K}_{\text{log utility}} \times \underbrace{\mathcal{K}_0^{2M}}_{\text{pos. balance}} \times \underbrace{\mathcal{K}_{\geq}^{2n+K+G}}_{\text{box + nonneg}}$$

Total slack dimension: $3K + 2M + 2n + K + G$. Number of exponential cones: K (typically 1–10).

The entire problem is specified by two sparse matrices (A , $P = 0$), two vectors ($\mathbf{b}$, $\mathbf{c}$), and a cone list. No callbacks, no oracles, no iteration.

2.6. Price Extraction

Clearing prices come from the duals of position balance constraints, identical to the LP:

$$p_{\text{YES},m} = |\lambda_{\text{YES},m}|, \quad p_{\text{NO},m} = |\lambda_{\text{NO},m}|$$

where λ are the dual variables (Clarabel's `solution.z` for the zero-cone block). When minting is active, $p_{\text{YES}} + p_{\text{NO}} = N$ by stationarity of the mint variable.

3. The Augmented-LP Insight

The conic reformulation is clean but treats Clarabel as a black box. A deeper perspective reveals that the EG problem is a *minimal perturbation* of the LP.

3.1. Hessian Structure

The EG objective $f(\mathbf{q}, \mathbf{s}) = \sum_k [B_k \ln(V_k) - s_k] + \sum_{j \notin \text{MM}} w_j q_j - (\text{minting cost})$ has Hessian:

$$\nabla_{q_i q_j}^2 f = - \sum_k \frac{B_k}{V_k^2} \cdot L_{k,i} \cdot L_{k,j}$$

where $L_{k,i}$ is the welfare weight of order i for MM k (L_i if $i \in \text{MM}_k$, else 0). In matrix form:

$$H_{\text{EG}} = - \sum_{k=1}^K \frac{B_k}{V_k^2} \cdot \ell_k \ell_k^T$$

This Hessian has rank at most K . Each $\ell_k \in \mathbb{R}^n$ has nonzeros only for MM k 's orders. For typical batches ($K = 1\text{--}10$ MMs, $n = 100\text{--}10000$ orders), the Hessian is trivially cheap to form and apply.

The cross-derivatives with cash variables s_k extend this cleanly. Define $\tilde{\ell}_k \in \mathbb{R}^{n+K}$ as ℓ_k extended with a 1 in position $n + k$. Then the full Hessian over $(\mathbf{q}, \mathbf{s})$ is:

$$H = - \sum_{k=1}^K \frac{B_k}{V_k^2} \cdot \tilde{\ell}_k \tilde{\ell}_k^T$$

Still rank- K , still negative semidefinite.

3.2. Newton's Method = Modified LP Interior Point

A standard LP interior-point solver minimizes the barrier subproblem $-\mathbf{c}^T \mathbf{q} + \mu \sum_i [-\ln(\bar{q}_i - q_i) - \ln(q_i)]$, solving the KKT system at each step:

$$\begin{pmatrix} D_{\text{LP}} & A^T \\ A & 0 \end{pmatrix} \begin{pmatrix} \Delta \mathbf{q} \\ \Delta \boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \end{pmatrix}$$

where $D_{\text{LP}} = \text{diag}(1/q_i^2 + 1/(\bar{q}_i - q_i)^2)$ is the barrier Hessian (diagonal, positive definite).

For EG, the KKT system becomes:

$$\begin{pmatrix} D_{\text{LP}} + \Sigma & A^T \\ A & 0 \end{pmatrix} \begin{pmatrix} \Delta \mathbf{q} \\ \Delta \boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \end{pmatrix}$$

The only change: D_{LP} gains the rank- K correction

$$\Sigma = -H_{\text{EG}} = \sum_k \frac{B_k}{V_k^2} \cdot \ell_k \ell_k^T$$

which is positive semidefinite (so the system stays positive definite).

3.3. Efficient Solve via Woodbury

The normal equations (Schur complement) are:

$$A(D_{\text{LP}} + \Sigma)^{-1} A^T \Delta \lambda = r$$

By Sherman–Morrison–Woodbury:

$$(D_{\text{LP}} + LL^T)^{-1} = D^{-1} - D^{-1}L(I_K + L^T D^{-1}L)^{-1}L^T D^{-1}$$

where $L \in \mathbb{R}^{n \times K}$ stacks the scaled $\sqrt{B_k}/V_k \cdot \ell_k$ vectors. The inner matrix $I_K + L^T D^{-1}L$ is $K \times K$ — trivially invertible. Each Newton step costs:

- $O(n)$: invert the diagonal D (same as LP)
- $O(nK)$: form $D^{-1}L$ and the $K \times K$ inner product
- $O(K^3)$: invert the $K \times K$ inner matrix

For $K \ll n$, this is dominated by $O(n)$ — **the same cost as one LP interior-point iteration.**

3.4. Iteration Count

Self-concordant barrier theory: an interior-point method converges in $O(\sqrt{\nu})$ iterations where ν is the barrier parameter (number of inequality constraints). The EG problem has the same constraints as the LP plus K log terms — but these K log terms are themselves self-concordant barriers. So $\nu_{\text{EG}} = \nu_{\text{LP}} + K$, and the iteration count is $O(\sqrt{n+K}) \approx O(\sqrt{n})$.

The EG solver has the same asymptotic complexity as the LP solver.

4. Performance Analysis

4.1. Problem Dimensions (Typical)

Parameter	Symbol	Small	Large
Orders	n	100	10,000
Markets	M	20	200
Groups	G	5	50
Market makers	K	2	10
Variables	d	~ 130	$\sim 10,300$
Constraints		~ 50	~ 500
Exp. cones		2	10

4.2. Expected Speedup

Method	Solves	Per-solve cost	Total
Frank-Wolfe (current)	25 LPs	O_{LP}	$25O_{\text{LP}}$
Clarabel (conic)	1	$\sim 3\text{--}5O_{\text{LP}}$	$3\text{--}5O_{\text{LP}}$
Augmented LP (custom)	1	$\sim 1.1O_{\text{LP}}$	$\sim 1.1O_{\text{LP}}$

- **Frank-Wolfe** (current `eg_solver.rs`): 25 iterations, each solving a full LP via HiGHS. Total: $25 \times O_{\text{LP}}$.
- **Clarabel**: Generic conic interior-point solver. Handles exponential cones natively but carries overhead vs. LP-specialized solvers (general-purpose KKT factorization, no LP presolve). Estimated $3\text{--}5 \times$ per-solve cost of HiGHS. Net speedup: **5–8×**.
- **Augmented LP**: Custom interior-point method that reuses the LP’s constraint structure and adds the rank- K Hessian correction. Per-iteration cost identical to LP interior point. Net speedup: **$\sim 20\times$** , approaching LP parity.

The Clarabel path is straightforward to implement (days of work). The augmented-LP path requires a custom interior-point solver (weeks) but achieves near-LP speed.

5. Implementation Plan

5.1. Phase 1: Clarabel (immediate)

1. Add `clarabel` dependency (feature-gated behind `eg-conic`).
2. Build the conic program from the same data structures as `build_and_solve_lp()`. Reuse `precompute_coefficients`, `collect_markets`, etc.
3. Construct sparse A matrix: stack exponential cone rows, position balance rows, box constraint rows. Build $\mathbf{b}$ and $\mathbf{c}$ vectors. Pass cone specification.
4. Extract primal fills ($\mathbf{q}$) and dual prices ($\boldsymbol{\lambda}$) from `solver.solution`.
5. Round fills to integers, normalize prices, run `trim_mm_budget_overflows`, create arb orders — same post-processing pipeline as current solvers.
6. Validate: compare fills and welfare against Frank-Wolfe EG on all existing test cases.

5.2. Phase 2: Augmented LP (future)

1. Implement a primal-dual interior-point solver for the LP constraint structure (LDL^T or Cholesky on the normal equations $AD^{-1}A^T$).
2. Add the rank- K Hessian correction via Woodbury identity.
3. Implement Mehrotra predictor-corrector for practical iteration efficiency.
4. This gives a single solver for both LP ($K = 0$) and EG ($K > 0$), unifying `lp_solver.rs` and `eg_solver.rs`.

6. Extension: Smoothed Minting Cost

The companion paper uses the smoothed minting cost $C_b(\mathbf{D}) = b \ln \sum_s \exp(D_s/b)$, which differs from the linear minting cost ($b \rightarrow 0$ limit) used in `lp_solver.rs`. The smoothed cost is also expressible via exponential cones.

The epigraph $C_b(\mathbf{D}) \leq \tau$ decomposes as:

$$\exists u_s \geq 0 : \sum_s u_s \leq 1, \quad (D_s/b - \tau/b, \quad 1, \quad u_s) \in \mathcal{K}_{\text{exp}} \quad \forall s \in \mathcal{S}$$

This adds $|\mathcal{S}|$ exponential cones — one per joint state. For a single binary market ($|\mathcal{S}| = 2$), trivial. For a decomposed component with $|\mathcal{S}| = 2^{10} \approx 1000$ states, still tractable.

The smoothed formulation is useful for:

- **Theoretical consistency** with the companion paper’s analysis
- **Better conditioning**: the log-sum-exp gradient is smooth, avoiding the LP’s vertex-hopping and degeneracy
- **Price uniqueness**: smoothed minting gives unique prices even at non-generic demand (no degenerate duals)

The practical solver should support both: linear minting (fast, matches existing solvers) and smoothed minting (robust, matches theory). The conic reformulation handles both — just toggle whether to include the $|\mathcal{S}|$ extra exponential cones.

7. Summary

Property	Frank-Wolfe	Clarabel	Augmented LP
Quasi-linear (s_k)	No	Yes	Yes
Single solve	No (25 LPs)	Yes	Yes
Price extraction	Projection LP	Dual vars	Dual vars
Implementation	Exists	Days	Weeks
Speedup vs. FW	$1\times$	$5\text{--}8\times$	$\sim 20\times$
Speedup vs. LP	$0.04\times$	$0.2\text{--}0.3\times$	$\sim 0.9\times$

The key insight: the EG clearing problem is not a “hard nonlinear program” — it is an LP with K logarithmic terms, where K (number of MMs) is tiny. Interior point methods handle this with negligible overhead beyond the LP. The Frank-Wolfe approach pays for 25 LP solves to approximate what one Newton-based solve gives exactly.

Design note for the Sybil matching engine. Companion to Prediction Markets Are Fisher Markets and Decomposed Combinatorial Clearing via Fisher Market Budget Allocation (2026).